Digital Literacy & Online Safety

Full Course Report — All 5 Tasks

Sri Vinissrith Avasarala
Roll No. 25BCY10273
B.Tech — Computer Science Engineering (Cyber Security & Digital Forensics)
School of Computing and Artificial Intelligence (SCAI)
VIT Bhopal University

March 2026

Task 1: Platforms Used — HackerRank & Google Forms

1.1 Overview

For Task 1, two digital platforms were explored: HackerRank for programming practice and Google Forms for knowledge assessment. Both tools were chosen for their relevance to academic and professional development in the field of Computer Science and Cyber Security.

1.2 HackerRank

Platform Purpose

HackerRank is an open, competitive programming platform that supports practice in Python, Java, C/C++, and numerous other languages through structured challenges categorised by difficulty and topic.

Activity Completed

The introductory Python challenge 'Say Hello, World! With Python' was completed successfully. The submission passed all test cases, earning 5.00 points and bringing the total Python score to 5/35 (14% progress toward the first star badge). The console confirmed a Success status for Test Case 0 with the expected output Hello, World!

Academic Value

HackerRank provides a structured, gamified pathway to develop programming proficiency. As a Cyber Security and Digital Forensics student, mastering languages such as Python (for scripting and automation), C/C++ (for low-level security analysis), and Java (for enterprise application security) is fundamental. The platform's leaderboard and badge system also encourages consistent practice, which is essential for building long-term technical skill.

1.3 Google Forms

Activity Completed

A Digital Literacy Awareness Quiz was designed and published using Google Forms. The quiz carried a total of 25 points and was configured to automatically collect respondents' email addresses. The quiz covered four distinct areas of digital safety:

- Password security — identifying the most secure password creation method
- Terms & Conditions awareness — understanding the legal implications of clicking 'Agree'
- Public Wi-Fi safety — evaluating the risk of banking on HTTPS sites over public networks
- Phishing recognition — responding appropriately to a suspicious email from 'support@paypa1.com'
- Digital Footprint — a short-answer question asking respondents to define the concept and give a personal example

The quiz was attempted by the creator on 29 March 2026 at 18:39:17, achieving a score of 20 out of 25. The response sheet, stored in Google Sheets (Form_Responses tab), confirmed the submission with the following notable answers:

- Password question: A Long Passphrase with mixed Characters (e.g. Mango$Sky99/River!) — Correct
- Terms & Conditions: FALSE — Correct (agreeing to T&Cs does not prevent misuse)
- Public Wi-Fi: FALSE — Correct (HTTPS does not fully protect data on public networks)
- Phishing email: Check the Sender's address carefully — Correct

Academic Value

Designing a quiz requires a thorough understanding of the subject matter. The process of crafting plausible but incorrect distractors for each question deepened personal knowledge of digital safety concepts. Google Forms also demonstrates how data collection and survey tools function in professional and research contexts, which is directly relevant to academic assessments and fieldwork in Cyber Security.

1.4 Reflection

HackerRank and Google Forms serve complementary academic purposes: the former builds technical coding skills progressively, while the latter develops assessment design and digital literacy awareness. Both will continue to play a role in academic growth — HackerRank as a daily practice environment and Google Forms as a tool for peer-testing and knowledge reinforcement.

Task 2: Professional Email Writing

2.1 Overview

Task 2 focused on developing professional written communication skills in a digital context. Two formal emails were drafted: a request for an assignment deadline extension addressed to a professor, and an internship application addressed to a company's hiring team. Both templates follow the conventions of professional academic and corporate correspondence.

2.2 Email 1 — Assignment Deadline Extension Request

Context & Purpose

This email is addressed to a course professor requesting additional time to complete an assignment. Such communication is a common professional necessity in both academic and workplace environments, requiring a careful balance between honesty, courtesy, and brevity.

Key Elements Included

- A clear, formal subject line identifying the course, code, and purpose of the email
- A polite opening acknowledging the professor's time
- A concise, honest explanation of the circumstances that caused the delay
- A specific proposed new submission date, rather than a vague request
- A commitment to completing the work within the extended timeframe
- A formal sign-off with full student details including Student ID and course information

Structural Observation

The email avoids over-explaining or offering excessive justification. The proposed new deadline is stated specifically rather than left open-ended, which demonstrates respect for the professor's time and makes the request easier to process. The tone remains formal and avoids any casual language, which would be inappropriate in an academic setting.

2.3 Email 2 — Summer Internship Application

Context & Purpose

This email is intended for a company coordinator or hiring team and serves as a cold-outreach internship application. Unlike a standard cover letter, this format must be self-contained and immediately engaging, as it competes with a high volume of similar applications in a professional inbox.

Key Elements Included

- A subject line that is informative and includes the student's name and institution for easy identification
- A brief self-introduction with degree programme and university in the first paragraph
- A specific reason for interest in the company, referencing a genuine area of admiration rather than using generic language

- A concise list of relevant skills with reference to supporting experience such as academic projects or extracurricular activities
- Clear availability dates and openness to multiple roles
- Mention of attached documents (resume, transcript, portfolio) to direct the recruiter's next steps
- Optional LinkedIn and GitHub profile URLs to demonstrate professional online presence

Structural Observation

The internship application email is structured in four focused paragraphs — introduction, interest and motivation, skills and evidence, and a call to action. This mirrors the structure of professional cover letters while remaining appropriately concise for email communication. Personalisation details (company name, specific initiative, exact skill set) are marked with placeholders to emphasise that the email must be tailored for each individual application — a critical best practice in professional outreach.

2.4 Reflection

Professional email writing is a foundational skill for any university student, and particularly for those pursuing careers in technology and cyber security, where clear and precise communication is as important as technical ability. Drafting these templates reinforced the importance of structure, tone, and specificity in written professional communication.

Task 3: Cybersecurity Case Study — The Google & Facebook Phishing Scam (2013–2015)

3.1 Overview

Task 3 involved researching a real-world cybersecurity incident to understand how phishing attacks are executed at scale, even against the world's most technologically sophisticated organisations. The chosen case — the defrauding of Google and Facebook by Lithuanian national Evaldas Rimasauskas between 2013 and 2015 — is one of the most significant and instructive corporate phishing cases in history.

3.2 What Is Phishing?

Phishing is a cyberattack technique in which criminals impersonate trusted entities — such as banks, government agencies, vendors, or colleagues — through emails, messages, or fake websites, with the goal of tricking victims into revealing sensitive information including passwords, financial credentials, or login details. It is one of the most prevalent and economically damaging forms of cybercrime globally.

3.3 How the Attack Was Executed

Rimasauskas orchestrated the attack over approximately two years using a method known as Business Email Compromise (BEC) — a sophisticated variant of phishing targeting corporate finance processes:

Step 1	Rimasauskas registered a fraudulent company in Latvia, deliberately named to impersonate Quanta Computer — a legitimate Taiwanese hardware manufacturer that both Google and Facebook used as a regular vendor.

Step 2	He crafted and sent professionally formatted fraudulent invoices via email to finance department employees at both technology companies.

Step 3	The emails were indistinguishable from legitimate vendor communications — they used correct branding, familiar vendor names, and plausible transaction amounts that matched existing business relationships.

Step 4	Finance employees, seeing nothing suspicious, approved the invoices and wired payments totalling over $100 million USD to bank accounts in Latvia and Cyprus controlled by Rimasauskas.

Step 5	The funds were rapidly laundered across multiple international accounts before either company detected the fraud.

3.4 Consequences

Both Google and Facebook eventually recovered a significant portion of the lost funds following the FBI's arrest of Rimasauskas in Lithuania in 2017. He was extradited to the United States, where he pleaded guilty to wire fraud in 2019. Beyond the immediate financial loss of over $100 million, both companies suffered considerable reputational embarrassment — a reminder that no organisation is immune to social engineering, regardless of technical sophistication. Both firms subsequently overhauled their vendor verification and payment authorisation processes.

3.5 Personal Reflection

The most striking aspect of this case is the sheer financial scale of the fraud: over 100 million US dollars lost by two of the world's largest and most technically capable companies. The attack succeeded not by breaking any encryption or exploiting a software vulnerability, but by exploiting human trust and standard business processes — a reminder that the human element is often the weakest link in any security chain.

As a result of studying this case, one specific personal habit change has been identified: always carefully examining the sender's email address and verifying URLs before clicking any link in an email, particularly when the message involves a financial transaction, account verification, or urgent request. This habit of deliberate sender verification is one of the simplest and most effective defences against phishing.

3.6 Key Lessons

- Phishing does not require technical hacking skills — social engineering alone can defraud billion-dollar corporations
- Vendor verification procedures must include multi-layer confirmation before any wire transfer is approved
- A two-person review policy for financial authorisations significantly reduces the risk of BEC fraud
- Employee training in recognising suspicious emails is not optional; it is a critical security control
- The sender address and domain of any financial email must be independently verified, not assumed correct from display name

Task 4: Checklists — Social Media Responsibility & Online Safety

4.1 Overview

Task 4 involved the creation of two practical checklists aimed at college students: one focused on responsible social media behaviour and the other on personal online safety and cybersecurity hygiene. Both documents were designed to be actionable, concise, and relevant to the specific digital risks faced by students in India.

4.2 Checklist 1 — Responsible Social Media Use

This checklist presents six Do's and six corresponding Don'ts structured as a two-column reference guide:

DO	DON'T
Think before posting — ask if a future employer, professor, or family member would be comfortable seeing it.	Don't overshare personal information such as real-time location, home address, phone number, or daily schedule.
Regularly review and update privacy settings to control who can see your profile and tagged content.	Don't engage in or amplify cyberbullying, callout posts, or content intended to humiliate others.
Verify information before sharing — cross-check news and claims with credible sources to avoid spreading misinformation.	Don't post under the influence — impaired judgment frequently leads to posts that cause lasting regret.
Use social media intentionally — set time limits and take breaks to protect mental health and academic focus.	Don't publicly criticise your college, professors, or classmates; even 'private' venting can be screenshotted.
Maintain a positive digital footprint — share achievements, projects, and interests that contribute to a professional personal brand.	Don't ignore red flags from strangers online, especially those who escalate quickly or request favours.
Give proper credit for quotes, images, and ideas you repost or reference to respect intellectual property.	Don't let social media replace real-life friendships and campus experiences.

4.3 Checklist 2 — Online Safety for College Students in India

This checklist targets ten critical cybersecurity areas with actionable, India-specific guidance:

- **Passwords & Accounts:** Use a unique, strong password (12+ characters with symbols, numbers, and mixed case) for every account. Use a trusted password manager like Bitwarden or 1Password and never reuse passwords across platforms.
- **Two-Factor Authentication:** Enable 2FA on all accounts — email, Instagram, WhatsApp, and banking apps. Prefer an authenticator app (Google Authenticator) over SMS-based OTPs wherever possible.

- **Phishing Awareness:** Never click links in unsolicited emails or WhatsApp messages, even if they appear to originate from your bank, college, or a government body. Always type the official URL directly into your browser.
- **UPI & Payment Safety:** Never share your UPI PIN, OTP, or QR code with anyone — not even someone claiming to be from your bank or NPCI. Remember that receiving money never requires entering your PIN.
- **UPI Transaction Habits:** Set a daily UPI transaction limit and enable SMS transaction alerts. Before scanning any QR code, verify the registered merchant name — fraudsters replace legitimate QR codes at shops and petrol pumps.
- **Public Wi-Fi Safety:** Avoid accessing banking apps, email, or college portals on public Wi-Fi. If necessary, connect through a reputable VPN to encrypt your data.
- **App Permissions & Downloads:** Only download apps from the Google Play Store or Apple App Store. Regularly audit and revoke unnecessary microphone, camera, and location permissions.
- **Identity & Document Safety:** Never upload photos of your Aadhaar, PAN card, or college ID to unverified websites or WhatsApp groups. Use the masked Aadhaar (available at uidai.gov.in) when partial ID is sufficient.
- **Social Media Privacy:** Set all profiles to private. Avoid publicly posting your college name, hostel, hometown, and daily schedule together — this combination is sufficient for targeted fraud or stalking.
- **Stay Updated:** Subscribe to cybersecurity alerts from CERT-In (cert-in.org.in) — India's official cybersecurity agency. Staying informed about emerging scam trends is one of the most effective personal defences.

4.4 Case Study Connection — US Airways 2013 Tweet Incident

To complement the checklists, a secondary case study was researched: the 2013 US Airways tweet incident. A social media employee accidentally posted a graphic and deeply inappropriate image in a public reply to a customer complaint on Twitter. The post remained live for over an hour, exposing millions of followers to the content and triggering immediate global outrage.

This incident powerfully illustrates why the social media Do's and Don'ts checklist exists — the consequences of a single lapse in digital attention are magnified enormously on public platforms. Three safeguards that could have prevented the incident entirely: a two-person review policy before any public post goes live, clearer workflows separating reporting from posting, and mandatory social media training for all staff managing brand accounts.

4.5 Reflection

Creating these checklists required synthesising knowledge across digital safety, social behaviour, and India-specific payment systems (UPI). The research process reinforced that cyber threats are not purely technical — they exploit human habits and complacency. Designing practical, memorable checklists is itself a valuable skill in cybersecurity awareness communication.

Task 5: Professional Online Presence — GitHub, LinkedIn & Canva Presentation

5.1 Overview

Task 5 focused on establishing and articulating a professional digital identity across three platforms: GitHub for technical portfolio and version control, LinkedIn for professional networking and career development, and Canva for visual presentation of digital literacy knowledge. A Stack Overflow account was also created as part of the broader community engagement goal.

5.2 GitHub

What Is GitHub?

GitHub is an open-source software development platform built on Git, the industry-standard version control system. It allows developers to host code repositories, collaborate with others, track changes, review pull requests, and contribute to open-source projects. It serves as a public-facing technical portfolio that employers, peers, and the broader developer community can view and evaluate.

Account & Repository Created

A public GitHub account was created under the username sri25bcy10273-cmd. A personal repository named SriVinissrithAvasarala was initialised with a README.md file containing the following introductory profile:

> *"My Name is Sri Vinissrith Avasarla | Roll No. 25BCY10273 | Branch: B Tech Computer Science Engineering (Cyber Security and Digital Forensics) | School: SCAI | What I Hope to Learn: Properly use Git"*

The repository has 1 branch (main), 0 Tags, and 2 commits as of the date of submission. The repository link is: https://github.com/sri25bcy10273-cmd/SriVinissrithAvasarala/tree/main

Future Plans for GitHub

Over the next four years, the repository will be expanded to include academic projects, cybersecurity scripts, CTF challenge solutions, and coursework code. GitHub will serve as a live, evolving record of technical growth and a key professional asset when applying for internships and full-time roles in cyber security.

5.3 LinkedIn

What Is LinkedIn?

LinkedIn is a professional social networking platform focused on career development, job searching, academic showcasing, and professional networking. Unlike general social media, LinkedIn is specifically designed to connect students with alumni, recruiters, industry professionals, and peer academics in a formal, career-oriented context.

Profile Created

A LinkedIn profile was created for Sri Vinissrith Avasarala with the following current details:

Name	Sri Vinissrith Avasarala

University	VIT Bhopal University

Degree	B.Tech — Computer Science Engineering, Cybersecurity & Digital Forensics (2025–2029)

Location	Hyderabad, Telangana, India

Profile URL	https://www.linkedin.com/in/sri-vinissrith-avasarala-b9aa7a394/

The profile is currently in its early stages with 0 followers and 0 post impressions, reflecting a freshly created account. The 'Open to' status has been activated, signalling openness to internship and job opportunities.

Future Plans for LinkedIn

LinkedIn will be used to connect with VIT Bhopal alumni, particularly those working in cyber security, digital forensics, and related technology fields. The profile will be progressively enriched with certifications (such as CompTIA Security+, CEH, or Google Cybersecurity Certificate), project descriptions, academic achievements, and skills endorsements. The long-term goal is to build a network that facilitates internship placements and mentorship during the four-year degree.

5.4 Stack Overflow

A Stack Overflow account was also created under the username 'Deluge' (user ID: 32559325) as part of building a broader professional developer community presence. The account was newly created and has 0 answers, 0 questions, and no reputation points yet, consistent with a first-day account. Stack Overflow will serve as a platform for seeking programming help, contributing answers, and engaging with the developer community as coding proficiency grows.

5.5 Canva Presentation — Digital Literacy Topics

Tool Used

Canva (canva.com) was used to design a visual presentation summarising key digital literacy topics covered throughout the course. Canva is a browser-based graphic design tool offering a library of pre-designed templates suitable for presentations, infographics, posters, and social media content.

Design Choices

A tan-coloured slideshow theme was selected from Canva's pre-designed templates. The warm neutral tone conveys professionalism while remaining visually approachable. The presentation covers three core topics across multiple slides:

- Useful Digital Tools for Students: Notion AI (for note-taking and AI-assisted study), Google Drive (for cloud storage and document collaboration), and To-Do List apps (for academic task management)
- Safe Internet Practices: Strong unique passwords, enabling Two-Factor Authentication, and using HTTPS websites exclusively
- Professional Online Presence: Googling yourself periodically to see what employers see, auditing public social media content, and engaging with industry content to build professional visibility

Reflection on Canva

The process of designing the presentation highlighted the value of visual communication in conveying digital literacy concepts. Pre-designed slide templates in Canva significantly reduce the time needed to create polished visual layouts, allowing focus to shift toward content quality rather than design mechanics. No particular difficulties were encountered; however, the range of available templates and the ease of customisation were noted as particularly useful features for future academic and professional presentations.

5.6 Reflection

Establishing a professional digital presence across GitHub, LinkedIn, and Stack Overflow at the start of a four-year degree programme is a strategic decision. These platforms collectively form a digital portfolio that will grow in value as projects, certifications, and skills accumulate. The earlier this presence is established and maintained, the stronger the professional profile at the point of internship and job applications.

The Canva presentation task reinforced the importance of being able to communicate technical knowledge to non-technical audiences through clear, visually structured materials — a skill increasingly valued in the cyber security industry, where professionals must often brief stakeholders, write reports, and present findings in accessible formats.

Conclusion

This report documents the completion of five structured tasks designed to build digital literacy, professional online presence, cybersecurity awareness, and communication skills across multiple platforms and formats. The table below summarises the key outcomes:

Task	Activity	Key Outcome
1	HackerRank & Google Forms	Python Hello World completed (5 pts); Digital Literacy Quiz created and scored 20/25
2	Professional Email Writing	Two templates drafted: deadline extension request and internship application
3	Phishing Case Study	Google & Facebook scam ($100M) analysed; personal habit change identified
4	Social Media & Safety Checklists	Two checklists created covering responsible social media use and India-specific cybersecurity hygiene
5	GitHub, LinkedIn & Canva	GitHub repository and LinkedIn profile created; Canva presentation on digital literacy topics completed

Across all five tasks, a consistent theme emerged: effective participation in the digital world requires both technical competence and behavioural discipline. Coding skills developed on HackerRank must be matched by an understanding of the threats these systems face, as studied through the phishing case. Professional profiles on GitHub and LinkedIn must be managed with the same care as the social media guidelines recommend. And the ability to communicate clearly — whether in a formal email, a checklist, or a visual presentation — is as important as any individual technical skill.

This foundation, established in the first semester of a four-year degree in Cyber Security and Digital Forensics, provides a strong base for continued academic and professional development.